

Portfolio Milestone Module 4: Intelligent Search for CubeSat Telemetry

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The “Portfolio Milestone Module 2: Use-Case Scenario Proposal” (Ricciardi, 2026) proposes a hybrid CubeSat telemetry anomaly-detection system in which intelligent search will support two complementary tasks: retrieving relevant historical evidence and generating an efficient diagnostic investigation plan. First, I would use a k-nearest-neighbor similarity search over standardized telemetry feature vectors to compare a newly selected OPS-SAT segment with labeled segments in the OPSSAT-AD dataset and return the most similar normal and anomalous cases, providing the operator with concrete historical context for the neural network’s anomaly probability (Ruszczak et al., 2025). Second, after the Artificial Neural Network (ANN) determines how likely the satellite data is to show a problem, the A* symbolic planner will use that result and the system’s diagnostic rules to select the appropriate checks and arrange them in an efficient order. The planner will represent the current telemetry evidence, triggered rules, and completed checks as a state; each simulated diagnostic test as an action; and a predefined diagnostic stopping condition, such as completion of all mandatory checks for the leading fault hypothesis, as the goal. A* is appropriate because it evaluates each candidate path with $f(n) = g(n) + h(n)$, where $g(n)$ records the accumulated cost of the checks already selected and $h(n)$ estimates the remaining cost, combining the path-cost awareness of uniform-cost search with the goal direction of greedy search (CS188 Spring 2014, 2014). In this use case, action costs could represent simulated test duration, computational effort, or operator burden, while a conservative heuristic could estimate the minimum cost of the mandatory diagnostic stages that remain incomplete. With nonnegative action costs and a consistent heuristic that does not overestimate the remaining cost, A* graph search can return a least-cost investigation sequence while avoiding a purely greedy method’s tendency to favor an apparently promising but ultimately expensive

path. This predictive-to-prescriptive design follows Sharda et al.'s (2024) explanation that predictive results can be combined with optimization and heuristic models to guide planning and recommend decisions. Together, similarity search and A* will convert the ANN's classification into explainable historical evidence and an ordered sequence of diagnostic actions while preserving the human satellite operator's authority over the final decision.

References

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